

Weak positive-density blocks imply type-2 disjoint unboundedness

A partial answer to the question in arXiv:1812.03824

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11 August 2026

Abstract

Kostić asks whether the weak-to-strong implication in Proposition 8 of Bernardes–Bonilla–Müller–Peris extends to disjoint distributionally unbounded vectors of types 1 and 2. We settle the type-2 case affirmatively on every Banach space. A positive-density block on which at least one member of a finite operator tuple is large contains, along a subsequence, a fixed component large on a positive fraction. The ordinary one-operator theorem then gives an upper-density-one unbounded orbit for that component, which is automatically type 2 for the tuple. We also settle type 1 when the space is finite dimensional, and for finite tuples of normal operators on a complex Hilbert space: the weak condition makes every component non-power-bounded, and outside finitely many proper stable subspaces all component orbits diverge along the full sequence. The unrestricted infinite-dimensional type-1 case remains open.

1 Source question and notation

The source is M. Kostić, *Disjoint distributional chaos in Fréchet spaces*, arXiv:1812.03824v4 [1]. On PDF page 23, after extending the density-one block criterion to the four disjoint types, it records the following missing implication:

It is worth noting that a simple application of [11 Proposition 8] yields several equivalent conditions for the existence of a d -distributionally unbounded vector of type 3 or 4 for any sequence of linear continuous operators $T_1 \in L(X), \dots, T_N \in L(X)$ on a Banach space X . In the present situation, we do not know how to reconsider the implication (i)' $\Rightarrow$ (i) of this statement for d -distributionally unbounded vectors of type 1 or 2 for orbits of linear continuous operators on Banach spaces (all remaining parts of this proposition hold in our framework).

Let X be a Banach space over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ and let $T_1, \dots, T_N \in \mathcal{L}(X)$, where $N < \infty$. For $x \in X$ and $\varepsilon > 0$, put

$$E_k^\cup(x, \varepsilon) = \{1 \leq n \leq k : \max_{1 \leq j \leq N} \|T_j^n x\| > \varepsilon\},$$

$$E_k^\cap(x, \varepsilon) = \{1 \leq n \leq k : \min_{1 \leq j \leq N} \|T_j^n x\| > \varepsilon\}.$$

The union gauge is equivalent, up to the fixed factor N , to the sum gauge in the source's definition of type 2.

For $\tau \in \{\cup, \cap\}$, call the following the weak and strong block conditions:

$$\begin{aligned} (W_\tau) : \quad & \exists \varepsilon, \eta > 0, \ x_r \rightarrow 0, \ k_r \uparrow \infty, & |E_{k_r}^\tau(x_r, \varepsilon)| &\geq \eta k_r; \\ (S_\tau) : \quad & \exists \delta > 0, \ z_r \rightarrow 0, \ m_r \uparrow \infty, & \frac{|E_{m_r}^\tau(z_r, \delta)|}{m_r} &\longrightarrow 1. \end{aligned}$$

These are precisely the tuple analogues of conditions (i)' and (i) under discussion. For one operator, their equivalence is Proposition 8 of Bernardes–Bonilla–Müller–Peris [2].

2 Proof intuition

Type 2 uses a union: at every counted time it is enough that one component is large. Since the tuple is finite, a fixed component is responsible for a positive fraction along an infinite subsequence. The existing one-operator theorem upgrades that component from positive density to upper density one. No synchronization of different orbits is needed.

Type 1 uses an intersection, so the same reduction is unavailable. It does work whenever non-power-boundedness already forces a large common supply of vectors whose full orbits diverge. Finite-dimensional operators and normal Hilbert-space operators have exactly this spectral property: their nondivergent vectors lie in proper closed linear subspaces. A finite union of such subspaces cannot fill the space.

3 The type-2 implication on arbitrary Banach spaces

Theorem 1 (Full type-2 weak-to-strong implication). *For every finite tuple $T_1, \dots, T_N \in \mathcal{L}(X)$ on an arbitrary Banach space,*

$$(W_\cup) \implies (S_\cup).$$

More strongly, there are an index j_0 , a vector $x \in X$, and a set $A \subseteq \mathbb{N}$ with $\overline{\text{dens}}(A) = 1$ such that

$$\lim_{n \in A} \|T_{j_0}^n x\| = \infty.$$

Consequently x is a disjoint distributionally unbounded vector of type 2 for the tuple. The set of all type-2 vectors is residual.

Proof. Assume $(W_\cup)$. For each r and j , let

$$F_{r,j} = \{1 \leq n \leq k_r : \|T_j^n x_r\| > \varepsilon\}.$$

Since $E_{k_r}^\cup(x_r, \varepsilon) = \bigcup_{j=1}^N F_{r,j}$, there is an index j_r with

$$|F_{r,j_r}| \geq \frac{\eta}{N} k_r. \tag{1}$$

After passing to a subsequence, $j_r = j_0$ is constant. The sequence $x_r \rightarrow 0$, the increasing horizons (after another subsequence if necessary), and (1) are condition (i)' of Proposition 8 in [2] for the single operator T_{j_0} , with density constant η/N .

That proposition gives a vector x and a set A of upper density one such that $\|T_{j_0}^n x\| \rightarrow \infty$ along A . It also gives the equivalent strong block condition for T_{j_0} . Since

$$\max_j \|T_j^n y\| \geq \|T_{j_0}^n y\|$$

for every y and n , the same strong blocks establish $(S_\cup)$, and

$$\sum_{j=1}^N \|T_j^n x\| \geq \|T_{j_0}^n x\| \longrightarrow \infty \quad (n \in A).$$

This is type-2 disjoint distributional unboundedness.

Proposition 8 also says that the set of distributionally unbounded vectors for T_{j_0} is residual. It is contained in the set of type-2 vectors for the tuple, so the latter set is residual as well. $\square$

4 Two full type-1 subcases

Lemma 2 (Spectral common-divergence principle). *Suppose that $T_1, \dots, T_N$ are all non-power-bounded and either*

1. *X is finite dimensional, or*
2. *X is a complex Hilbert space and every T_j is normal.*

Then there is $x \in X$ such that

$$\|T_j^n x\| \longrightarrow \infty \quad \text{for every } j = 1, \dots, N.$$

Proof. In finite dimension, Jordan normal form gives, for each non-power-bounded T_j , a proper linear subspace M_j containing every vector whose orbit does not tend to infinity. Explicitly, M_j consists of the full generalized eigenspaces for eigenvalues of modulus < 1 , the eigenspaces (but not higher Jordan levels) for eigenvalues of modulus 1, and no nonzero component from generalized eigenspaces of modulus > 1 . Outside M_j , a fastest nonzero Jordan component grows exponentially, or polynomially in a nontrivial unit-modulus Jordan block. The same description follows by complexification over $\mathbb{R}$. Since finitely many proper linear subspaces cannot cover X , choose $x \notin \bigcup_j M_j$.

In the normal Hilbert-space case, non-power-boundedness means $\|T_j\| > 1$. Let

$$M_j = \mathbf{1}_{\{|z| \leq 1\}}(T_j)X.$$

This is a proper closed subspace. If $x \notin M_j$, its spectral measure $\mu_{j,x}$ gives positive mass to $\{|z| > 1\}$, and the spectral theorem gives

$$\|T_j^n x\|^2 = \int |z|^{2n} d\mu_{j,x}(z) \longrightarrow \infty$$

by monotone convergence on that set. Each $X \setminus M_j$ is open dense, so their finite intersection is nonempty. Any x in it works. $\square$

Theorem 3 (Type 1 in spectral settings). *If either hypothesis of Lemma 2 holds, then*

$$(W_\cap) \implies (S_\cap).$$

Indeed, there is a vector whose T_j -orbits all diverge along the full sequence $\mathbb{N}$; hence it is type-1 disjoint distributionally unbounded.

Proof. Condition $(W_\cap)$ implies that every T_j is non-power-bounded. If $\sup_n \|T_j^n\| = C_j < \infty$, then for all sufficiently large r ,

$$\|T_j^n x_r\| \leq C_j \|x_r\| < \varepsilon \quad (n \geq 0),$$

contradicting the positive cardinality required by $(W_\cap)$. Lemma 2 therefore supplies x with all component orbits diverging.

To obtain the block form explicitly, put $z_r = x/r$. Choose a_r so that $\min_j \|T_j^n z_r\| > 1$ for every $n \geq a_r$, and then choose increasing $m_r \geq ra_r$. Thus

$$\frac{|E_{m_r}^\cap(z_r, 1)|}{m_r} \geq 1 - \frac{a_r}{m_r} \geq 1 - r^{-1} \longrightarrow 1.$$

This is $(S_\cap)$. $\square$

5 Scope, checks, and novelty guardrail

Theorem 1 resolves the source’s missing implication completely for type 2; it does not require commutativity, separability, or any spectral hypothesis. The finiteness of the tuple is essential to the pigeonhole constant. Theorem 3 is a genuine subcase only: for arbitrary infinite-dimensional nonnormal operators, individual upper-density-one unbounded sets need not synchronize, and non-power-boundedness does not force full-orbit divergence on a dense open set.

Eight focused proof attempts were recorded. The unsuccessful unrestricted type-1 routes included intersecting individual density-one sets, using a direct-sum operator on X^N , a minimum-of-norms Baire gauge, tensor-power linearization, and attempts to synchronize the time-shift step in the proof of Proposition 8. Each loses either invariance of the diagonal or control of a common time set. This is the precise remaining obstruction, not an unproved step in the theorems above.

On 11 August 2026, bounded novelty searches covered the run’s four cheap indexes; the exact arXiv id and title; the quoted phrase “implication (i)’ $\Rightarrow$ (i)” ; “distributionally unbounded vectors of type 1 or 2”; and later work on disjoint distributional chaos and weighted translations. They found the source and later applications of its definitions, but no later answer to this implication or the type-2 pigeonhole reduction. This is a bounded novelty guardrail, not a priority claim.

Human-review recommendation. Review as a substantial partial result. For Theorem 1, check that the source’s type-2 sum gauge is interchangeable with the maximum gauge for a finite tuple and that Proposition 8 is applied after fixing one component on a subsequence. For Theorem 3, check the finite-dimensional stable subspace description and the normal spectral projection argument.

References

- [1] M. Kostić, *Disjoint distributional chaos in Fréchet spaces*, arXiv:1812.03824v4 (2019).
- [2] N. C. Bernardes Jr., A. Bonilla, V. Müller, and A. Peris, *Distributional chaos for linear operators*, J. Funct. Anal. **265** (2013), 2143–2163, doi:10.1016/j.jfa.2013.06.019.